

Hệ điều hành nâng cao

Đồng thuận



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN



Bài toán đồng thuận – consensus

□ a set of processes must all agree on a decision

- a single commander sends a value to every node and all nodes reach agreement on this single value
- each node has an initial value, based on that, all nodes reach an agreement on a single value

□ requirements

- consistency: all processes agree on the same value at termination
- validity: the agreed-upon value must be some process's initial value
- termination: each process decides within a finite number of steps



Bài toán đồng thuận - Consensus

□ Impossibility -- Alice and Bob Example

- Alice sends Bob an email at 10:00am: "Let's meet at noon in Solaris Lab"
 - should Alice go at noon without an *ack*
- Bob sends back an *ack*
 - should Bob go (Alice may not go without receiving the *ack*)
- Alice sends back an *ack* for Bob's *ack*
 - should Alice go
- how can they make an agreed-upon decision without risking not meeting the other at noon

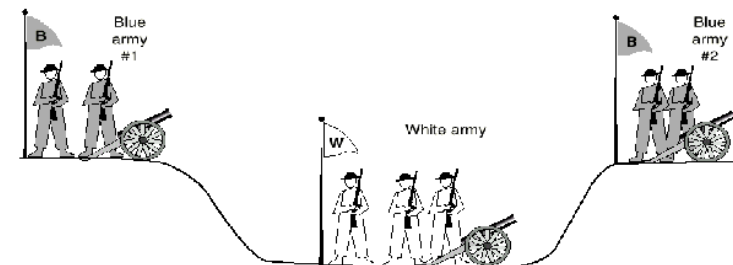


Fig. 6-13. The two-army problem.



Thuộc tính hệ thống

- ✗ processors: synchronous/asynchronous
 - + synchronous: for every $s+1$ steps taken by any processor, every other processor will have taken at least one step
 - + asynchronous example: Bob may get fired and no step will be taken
- ✗ communication delay: bounded/unbounded
 - + bounded: every message sent will arrive at the destination processor within t real time
 - + unbounded example: the email can reach its destination at any time
- ✗ message order: ordered/unordered
- ✗ message: point-point transmission/atomic broadcast
 - + atomic broadcast: a processor can send a message to all processors in one atomic step



Bầu cử leader – mạch vòng (LCR)

- Each process has a unique identifier
 - ▣ P_i 's identifier is i
- Assumptions
 - ▣ Synchronous system
 - ▣ The processes form a ring
- Goal
 - ▣ Try to identify the process with the minimal identifier as the leader



Bầu cử leader – mạch vòng (LCR)

- Election phase
 - ▣ Each process sends its identifier to its left neighbor
 - ▣ When P_i receives an identifier k
 - ▣ If $i < k$: ignore the message;
 - ▣ If $i > k$: forward the identifier k to its left neighbor;
 - ▣ If $i = k$: P_i 's identifier must have circulated across the entire ring and win the election
 - ▣ P_i sends selected(i) to its left neighbor
- Declare winning phase
 - ▣ When P_x receives the selected(i) message
 - ▣ If $i \neq k$: then forward selected(i) to its left neighbor
 - ▣ Else selected(i) has come back to P_i , stop



Bầu cử leader – mạch vòng (LCR)

- Message complexity: $O(n^2)$
- Time complexity: $O(n)$
- Can be modified to a broadcast based algorithm





Bầu cử leader-- Broadcast Based

□ Each process has a unique identifier

- P_i 's identifier is i

□ Assumptions

- Synchronous system
- Each process knows the identifier and IP of all other processes
 - Consider a closed group

□ Goal

- Try to identify the process with the maximal identifier as the leader



Bầu cử leader -- Broadcast Based

□ Election phase

- P_i sends its identifier i to all processes with a higher identifier
 - Wait for "Stop" responses, if one comes, then drop out
 - It means one of the processes with a higher identifier is alive
 - If none came, then P_i is the leader
 - Send a "selected(i)" message to all processes with a lower identifier
- When P_i receives an identifier k
 - If $i > k$: send back a "Stop" message and start its own election

□ Declare winning phase

- When P_x receives the selected(i) message
 - Set P_i as the leader

